SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Network Security & Cryptography: Symmetric/Asymmetric Ciphers & PKI

Department: CSE CS	**Course / Subject:** Network Security and Cryptography (CS505PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Classical encryption ciphers, DES/AES algorithms, Public-Key Cryptography (RSA, ECC), Hash functions (SHA-256), Digital Signatures, and Public Key Infrastructure.

Unit 1: Classical Ciphers & Symmetric Cryptography

Security attacks (Passive vs Active), Substitution ciphers, Transposition ciphers, Stream vs Block ciphers, Feistel cipher structure.

Unit 2: Modern Symmetric Block Ciphers

Data Encryption Standard (DES), Triple-DES, Advanced Encryption Standard (AES) rounds (SubBytes, ShiftRows, MixColumns, AddRoundKey), Modes of operation (ECB, CBC, CFB, CTR).

Unit 3: Asymmetric Cryptography & Key Exchange

Fermat's and Euler's theorems, Discrete logarithms, Diffie-Hellman Key Exchange, RSA algorithm mathematical proof, Elliptic Curve Cryptography (ECC).

Unit 4: Cryptographic Hash Functions & Digital Signatures

Message Authentication Codes (MAC), HMAC, SHA-256 and SHA-512 compression functions, Birthday paradox, Digital Signature Standard (DSS).

Unit 5: Network Security Protocols & Firewalls

Kerberos authentication, X.509 digital certificates, SSL/TLS handshake, IPsec (AH and ESP protocols), Firewalls (Packet filtering, Stateful inspection, Proxy).